



Contents lists available at ScienceDirect

Journal of Macroeconomics

journal homepage: www.elsevier.com/locate/jmacro

Forecasting recessions: the importance of the financial cycle

Claudio Borio, Mathias Drehmann^{*}, Fan Dora Xia¹

Bank for International Settlements, Centralbahnplatz 2, 4002 Basel, Switzerland

A B S T R A C T

Financial cycles can be important drivers of real activity, but there is scant evidence about how well they signal recession risks. We address this question, using a range of financial cycle measures. Unlike most papers, ours assesses forecasting performance not just for the United States but also for a panel of advanced and emerging market economies. We find that financial cycle measures have significant forecasting power both in and out of sample, even for a three-year horizon. Moreover, they outperform the term spread - the most widely used indicator in the literature - in nearly all specifications. These results are robust to different recession specifications.

1. Introduction

Predicting recessions has been a long-standing quest for market practitioners, policymakers and academics alike. As early as the first volume of *The American Economic Review*, Irving Fisher (1911) looked at developments in the “nation’s bank book” to forecast the likelihood of a contraction in the United States.² Following Estrella and Mishkin (1998), a consensus has emerged that the inverted yield curve, i.e. long-term bond yields below short-term interest rates, is among the best signals of impending recessions, if not the best one.

The notion that finance matters for the real economy has regained significant attention, especially after the Great Financial Crisis.³ Some authors even argue that all recessions in the United States since 1985 had financial origins (Ng and Wright (2013)).⁴ Despite this, research exploring how financial developments affect recession risks, i.e. the likelihood that a recession will occur in the near future, and whether they outperform the term spread as recession signal is scant. We fill this gap.

We explore in particular the importance of the financial cycle for predicting recession risks. This is not only of interest

This paper is an extension of “The financial cycle and recession risk”, BIS Quarterly Review, December 2018, by the same authors

^{*} Corresponding author.

E-mail addresses: claudio.borio@bis.org (C. Borio), mathias.drehmann@bis.org (M. Drehmann), dora.xia@bis.org (F.D. Xia).

¹ The authors would like to thank the editor and two helpful referees as well as Stijn Claessens, Ben Cohen, Mikael Juselius, Marco Lombardi, Hyun Song Shin and Kostas Tsatsaronis for helpful comments and Anamaria Illes for excellent research assistance. The views expressed in this article are those of the authors and do not necessarily reflect those of the BIS.

² Fisher’s starting point was the quantity theory of money, so he focused on bank deposits. But he also discussed the role of credit.

³ A large and growing body of work supports the link between finance and the real economy empirically. For one, there is substantial evidence that medium term credit and asset price developments are helpful leading indicators of financial crises (e.g. Borio and Drehmann (2009), Schularick and Taylor (2012), Detken et al. (2014)). Moreover, several studies have documented that credit booms weaken output in the medium run (e.g. Mian et al. (2017)), including by sapping productivity growth (Borio et al. (2016)). And some recent work has begun to study the impact of financial conditions on risks to growth (e.g. Adrian et al. (2018)). See Claessens and Kose (2018) for a review.

⁴ Oman (2019) shows that after 1990 all euro-area rescissions were preceded by a financial cycle boom.

academically, but also from a policy perspective as the concept of the financial cycle is underpinning financial stability analyses in many policy institutions.⁵

In the spirit of Minsky (1982) and Kindleberger (2000), the term “financial cycle” refers to the self-reinforcing interactions between asset prices, risk, risk taking, and financing constraints (Borio (2014)).⁶ To measure the financial cycle empirically, the literature has emphasised the role of joint, medium-term fluctuations - typically in the range of 10-20 years – in credit and asset prices as important interactions between credit, asset prices and the real economy play out at these frequencies along two dimensions. First, medium-term cycles of credit and asset prices are especially highly correlated with the medium-term GDP cycle (e.g. Runstler and Vlekke (2018) or de Winter et al. (2017)). Second, peaks in the (medium-term) financial cycle coincide with considerable financial stress or banking crises (e.g. Drehmann et al. (2012)), which in turn usher in deep and protracted recessions (e.g. Jordà et al. (2013)).

In this paper, we look at two financial cycle measures specifically. First, the “composite” financial cycle proxy as in Drehmann et al. (2012). This combines medium-term cycles in real credit, the credit-to-GDP ratio and real residential property prices. Second, the debt service ratio (DSR), defined as interest payments plus amortisation divided by GDP. Juselius and Drehmann (2020) show that the financial cycle can be described by the joint behaviour of leverage and the DSR. The DSR, in turn, is the key link to the real economy as it has a large negative impact on GDP growth (see also Drehmann et al. (2018)).

By focussing on the financial cycle, we expand on the few papers in the literature that assess the predictive power of individual variables that proxy some aspects of the financial cycle. For instance, the literature has found balance sheet conditions, property prices, credit growth or corporate credit spreads to be important (e.g. Ng (2012), Liu and Mönch (2016), Ponka (2017), Guender (2018) or Hwang (2019)).

As our aim is to better understand how financial cycles affect recession risks, we do not try to maximise overall forecast performance. We run simple panel probit models and do not do a horse race across all possible explanatory variables.⁷ But it is clear from the literature that using more elaborate methods such as time varying coefficient models, Bayesian model averaging or machine learning would boost forecast performance even further (e.g. Berge (2015) or Hwang (2019)).

We assess predictive performance of the financial cycle both in and out of sample. We rely on the area under the receiver operating characteristic (ROC) curve (AUC) to judge forecast performance (Berge and Jordà (2011)). Throughout, we compare the performance of the financial cycle with that of the term spread in light of the importance of the term spread as a benchmark even though its performance has shown to change over time (e.g. Chauvet and Potter (2002) or Hwang (2019)).

And in contrast to much of the literature, we assess the signalling power of financial cycle measures and the yield curve not only for the United States but also for an unbalanced panel of 16 advanced economies and nine emerging market economies (EMEs) from 1985 to 2017. This makes our results significantly more general. For instance, of the above-mentioned papers that look at variables related to the financial cycle, only Guender (2018) goes beyond the United States and considers a panel of European countries. And even the literature on the information content of the yield curve barely looks at EMEs; the exceptions are Mehl (2009), who examines the effect of the yield curve on growth; Öztürk and Pereira (2013), whose panel analysis includes EMEs; and several country-specific studies (e.g. Grabowski (2009) for Poland, Zulkhibri and Rani (2016) for Malaysia, and Paweenawat (2017) for Thailand)).

We find that financial cycle measures are very useful for gauging recession risks. They perform generally better than the term spread. When the competing variables are considered on a standalone basis, AUCs for financial cycle measures are statistically significantly higher and are significant even for a three-year horizon, for which the term spread is uninformative. When financial cycle measures and the term spread are included jointly in a probit model, they all remain statistically significant up to a two-year horizon. But combining information from the spread and financial cycle measures improves only marginally, and not significantly, AUCs and other evaluation metrics relative to specifications that include just financial cycle measures.

The high forecasting performance of the financial cycle measures also applies to the out-of-sample tests. Here, we carry out two exercises to assess the indicators’ predictive content in (quasi) real time. We examine first the effect of real-time data⁸ and fixed model parameters estimated using our first 10 years of data. We then assess the combined effect of real-time data and model parameters estimated recursively. In both exercises, the financial cycle measures retain statistically significant forecasting power.

The results apply to both advanced economies and EMEs. That said, the forecast performance of both the term spread and the financial cycle is generally weaker in EMEs, possibly reflecting much more volatile business cycles there. Across countries, a reason for the disappointing performance of the term spread is that for several economies the variable is “contaminated” by credit risk premia.

⁵ Discussions of the financial cycle feature regularly in financial stability reports or other publications of major central banks and international financial institutions including from the Bank of England, the Bank of Japan, the BIS, the ECB or the IMF.

⁶ Starting with Kiyotaki and Moore (1997) and Bernanke et al. (1999), the financial accelerator literature has emphasized the importance of financial constraints in amplifying macroeconomic fluctuations. And if constraints bind occasionally, crises periods with low growth and high volatility can emerge endogenously (e.g. Brunnermeier and Sannikov (2014)). Geanakoplos (2010) shows that the interaction between risk taking, asset prices and leverage can give rise to endogenous leverage cycles. Models of indebted demand (Mian et al. (2020)) may also explain why credit booms are followed by downturns.

⁷ A horse race exploring a large range of variables is also problematic from a data availability perspective. Several indicators such as credit spreads are not widely available for our panel of countries for a sufficiently long time series.

⁸ For the data, we would ideally use real-time releases. However, these are not available for the panel, so we use quasi-real-time information, i.e. we take the latest data vintage and calculate the financial cycle and normalise all variables using data only up to that point.

As a result, in some episodes, the yield curve steepened rather than flattened ahead of recessions, including in some periphery countries ahead of the 2011–12 euro sovereign debt crisis. By contrast, the financial cycle measures are immune to this problem.⁹ Consistently with this explanation, the term spread retains forecasting information for the United States as shown by the literature.

The performance of financial cycle proxies is robust regardless of whether we forecast the likelihood that the economy will be in a recession at a *specific* point in time several periods ahead – the literature standard – or whether the business cycle will turn *within* the next few periods. Following much of the literature (e.g. Rudebusch and Williams (2009)), in the main text we focus on the first approach – in our case one, two or three years ahead. As an alternative, we estimate the risk that the business cycle will turn from boom to bust within the next one, two or three years. This second type of exercise is much closer to the spirit of Irving Fisher, who had observed in 1911: “The exact date of such crisis (recession in his context), of course, it would be foolish to predict, but it if it occurs it would seem likely to occur between, say 1913 and 1916” (Fisher (1911), p 304).

The rest of the paper is structured as follows. In the second section, we briefly introduce the notion of the financial cycle and document how the nature of the business cycle, and its link with the financial cycle, have changed in the past 50 years. In the third section, we explain our methodology. In the fourth, we evaluate the performance of financial cycle proxies and compare it with that of the term spread based on full-sample information, i.e. *ex post*. In the fifth, we consider out-of-sample exercises, seeking to mimic the information policymakers have when assessing risks in real time, i.e. *ex ante*. In the sixth, we consider a different definition of recession risk as robustness check. Then we conclude.

2. The financial cycle and recession risk: a look at the data

Over the last decade, a growing literature has measured the financial cycle empirically relying on similar methods as developed by the business cycle literature. Initially, the literature has identified the financial cycle either by turning point analysis in the spirit of Burns and Mitchell (1946) (e.g. Claessens et al. (2012) or Drehmann et al. (2012)) or by band-pass filters (e.g. Aikman et al. (2015) or Drehmann et al. (2012)). But a range of more sophisticated methodologies has been employed such as unobserved component models (e.g. Galati et al. (2016) or Runstler and Vlekke (2018)), spectral analysis (e.g. Strohsal et al. (2019) or Schuler et al. (2020)) or wavelet tools (e.g. Verona (2016)).

Two important empirical stylised facts emerge from this literature.

First, the financial cycle is best identified by credit and asset-prices jointly. This is in line with the theoretical literature highlighting the importance of leverage (e.g. Geanakoplos (2010)). For asset prices, the role of residential property prices has been stressed in particular.¹⁰

Second and as discussed above, the financial cycle is typically much longer than the business cycle. Business cycles as traditionally measured have tended to last up to eight years, and financial cycles around 15–20 years since the early 1980s. The difference in length means that a financial cycle can span more than one business cycle. As a result, while financial cycle peaks tend to usher in recessions, not all recessions are preceded by such peaks.

In this paper, we rely on the financial cycle as in Drehmann et al. (2012) which captures both these features. We call it the “composite financial cycle”. To derive it, we first extract medium-term fluctuations with a band-pass filter and frequencies of 8 to 30 years from real credit, the credit-to-GDP ratio and real residential property prices. These are then combined into the composite financial cycle by simply averaging across these series.

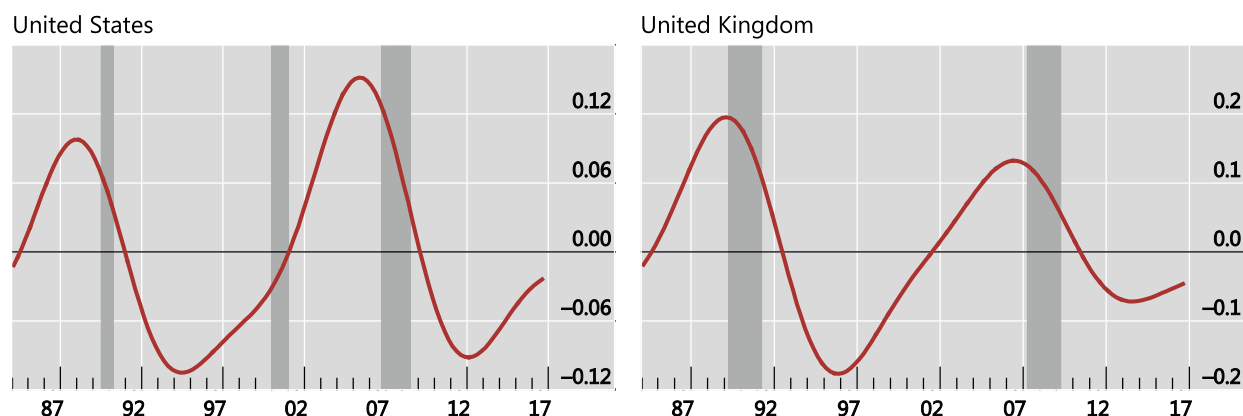
The focus on the financial cycle is supported by analysis in Annex 1, where we examine the forecasting performance of short and medium-term cycles in credit, the credit-to-GDP ratio and property price developments separately (as well as their annual growth rates). This shows that medium-term cycles in credit and residential property prices significantly outperform short-term cycles (i.e. cycles at business cycle frequency) in predicting recessions. Moreover, the composite financial cycle captures both the superior performance of the medium-term credit cycle for short (i.e. 1 year) and that of the medium-term property prices cycle for longer forecast horizons (i.e. 3 years). The importance of medium-term credit and asset price cycles for financial and real interactions is in line with the literature (e.g. Runstler and Vlekke (2018) or de Winter et al. (2017)). The limited forecasting performance of short-term cycles or annual growth rates is also not surprising given de Winter et al. (2017) find no empirical evidence for correlation between the short-term cycles of housing and GDP and only very little between the short-term cycles of credit and GDP.

In addition to the composite financial cycle, we assess the usefulness of the DSR. Based on a cointegrated VAR, Juselius and Drehmann (2020) show that the financial cycle can be described by the joint behaviour of leverage and the DSR. We focus on the DSR as the transmission from credit boom to the real economy arises via the effects of new borrowing on debt service, which in turn has a large negative effect on growth (see Drehmann et al. (2018)).

A first look at the relationship between the composite financial cycle proxy and recessions in the United States and the United Kingdom since the early 1980s corroborates the empirical patterns discussed above (Graph 1). Financial cycle booms took place ahead of recessions in the early 1990s and the late 2000s – both of which were associated with financial crises or periods of severe financial stress. At the same time, the shallow recession in the early 2000s in the United States did not coincide with a financial cycle peak: while

⁹ The zero (effective) lower bound and unconventional monetary policy may affect the term spread’s informational content to forecast recession risk, as the lower bound constrains short rates and unconventional monetary policy depresses the term premium in long rates. For a discussion, see Coroneo and Pastorello (2017), Fendel et al. (2018) and the minutes of the Federal Open Market Committee (March 2019) for examples.

¹⁰ Equity prices are less useful inputs. They exhibit a higher volatility at short-term frequencies than credit and property prices (as well as GDP). And they co-move little with these two series. Ideally commercial property prices would be considered but the data are generally too limited to do so.



The shaded areas represent recessions.

¹ Financial cycles are measured by the composite financial cycle proxy calculated from frequency-based (bandpass) filters capturing medium-term cycles in real credit, the credit-to-GDP ratio and real house prices.

Graph 1. Financial cycles tend to boom ahead of recessions¹

The shaded areas represent recessions.

¹ Financial cycles are measured by the composite financial cycle proxy calculated from frequency-based (bandpass) filters capturing medium-term cycles in real credit, the credit-to-GDP ratio and real house prices.

the economy slowed and equity prices tanked, the financial expansion – as measured by credit and property prices – continued, only to reverse a few years later, triggering the Great Recession. By contrast, in the United Kingdom, no actual recession took place in the early 2000s, so that the two recessions coincided with the two financial cycle peaks.

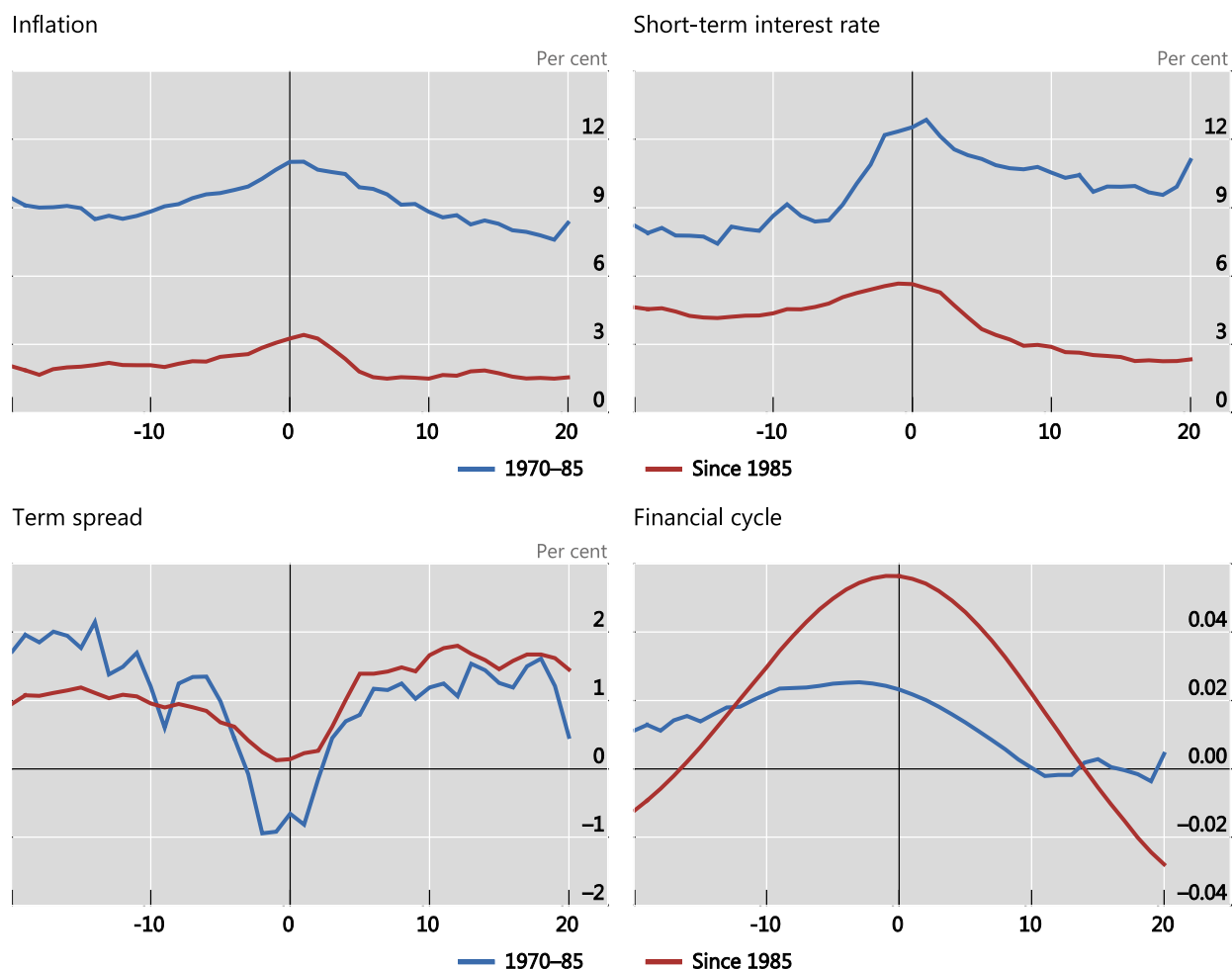
A longer-term cross-country perspective reveals an interesting change in the nature of recessions.¹¹ Graph 2 documents the behaviour of key variables in the five years around business cycle turning points in our sample of 16 advanced economies (vertical lines). In the period 1970–84 (blue lines), inflation and the short-term interest rate tended to increase by several percentage points ahead of cyclical peaks and term spreads to plunge and become highly negative (first three panels). At the same time, there was little sign of a financial boom (fourth panel). By contrast, since 1985 (red lines), around business cycle peaks inflation has been lower and remarkably stable, the short-term interest rate has increased only modestly and the term spread has narrowed far less. Correspondingly, strong financial cycle expansions have been very much in evidence. One could say that there has been a shift from inflation-induced¹² to financial cycle-induced recessions.

Why this is the case is less clear. But three changes deserve particular attention. First, financial markets were liberalised starting in the 1980s. Without sufficient prudential safeguards, this change is likely to have allowed greater scope for the self-reinforcing interactions at the heart of the financial cycle to play out. Second, starting roughly at the same time, inflation-focused monetary regimes became the norm. And the evolving thinking of central banks led them to gradually downplay the role of monetary and credit aggregates. This meant that central banks had little reason to tighten monetary policy if inflation remained low, even as financial imbalances built up. Finally, from the 1990s on, the entry of China and former communist countries into the world economy, alongside the international integration of product markets and technological advances, boosted global supply and strengthened competitive pressures, sapping labour's and firms' pricing power. Coupled with greater central bank credibility, this arguably made it more likely that inflationary pressures would remain muted even as expansions gathered pace. It also meant that financial booms could build up further and that a turn in the financial cycle, rather than rising inflation and the consequent monetary tightening, would trigger an economic downturn.¹³

¹¹ Interestingly, the recessions since the early 1980s have come to resemble those that were the norm under the pre-World War I classical gold standard (e.g. Huffman and Lothian (1984)) and in the run-up to the Great Depression in the 1920s (Eichengreen and Mitchener (2003)). This was the previous globalisation era; like today's, it was also characterised by price stability and a high degree of both trade and financial integration. See BIS (2018).

¹² Zarnowitz (1999) uses the term “central bank recession” to refer to the common view that recessions are always driven by monetary policy tightening.

¹³ For a discussion of changes in policy regimes and their implications for monetary and financial stability, see e.g. Borio and Lowe (2002) and Borio (2016).



¹ The horizontal axis denotes quarters around recessions in the business cycles, with the peak date set at zero (vertical lines). Lines show the median evolution across the advanced economies in our sample and events in the respective time period.

Graph 2. The changing nature of the business cycle¹

Average of the variables indicated over the selected period

¹The horizontal axis denotes quarters around recessions in the business cycles, with the peak date set at zero (vertical lines). Lines show the median evolution across the advanced economies in our sample and events in the respective time period.

3. Data and methodology

The previous graphical evidence is highly suggestive of the information that financial cycle proxies can convey about recession risk. In addition, it provides two indications for our more formal empirical analysis. For one, because of the changing nature of the business cycle, it makes sense to start in 1985. In addition, because financial cycles build up gradually, it is appropriate to focus on medium-term horizons in addition to the short-term ones common in the literature. We therefore aim to predict recessions up to three years ahead.

We use quarterly data for 16 advanced economies from 1985 to 2017.¹⁴ In general, we only include a country if all explanatory

¹⁴ Our sample comprises Australia, Belgium,* Canada, Finland,* France, Germany, Ireland,* Italy, Japan, the Netherlands,* Norway,* Spain, Sweden, Switzerland, the United Kingdom and the United States. For countries marked with an asterisk, we date business cycles using a business cycle-dating algorithm.

variables are available. A few countries therefore enter the sample later, mostly because of a lack of data for the term spread.¹⁵

To assess the broader validity of the results, we also look at quarterly data for nine EMEs.¹⁶ For these, data start in 1996 at the earliest and more often around 2000. One limiting factor is that government bond markets were less developed before then, constraining the sample for the term spread. In addition, property prices are scarcer for earlier dates, limiting the ability to compute the composite financial cycle proxy. In the case of EMEs, therefore, we do not use a homogeneous panel.¹⁷ While we still show results for the composite financial cycle proxy for comparability, these should be treated as indicative given the short time series.

We take the recession dates from the National Bureau of Economic Research (NBER) or the Economic Cycle Research Institute (ECRI). These rely on expert judgment based on the behaviour of several variables, such as output and employment. When such recession dates are not available, we rely on a standard business cycle-dating algorithm that identifies peaks and troughs in real GDP (Harding and Pagan (2002)). We do not consider degrees of intensity: either a recession occurs ($R=1$) or it does not ($R=0$). This, of course, means that countries with, on average, higher trend growth rates may experience fewer recessions but as many and equally sizeable slowdowns in growth. This is less of an issue for advanced economies.

For the main analysis, we follow most of the literature and predict whether the economy is in a recession h periods ahead ($R_{i,t+h} = 1$) based on the state of explanatory variables ($X_{i,t}$) at time t . Hence, in the panel probit setup we estimate:

$$\text{Prob}(R_{i,t+h} = 1 | X_{i,t}) = \Phi(\alpha_h + \beta_h' X_{i,t}) \quad (1)$$

To judge forecast performance, we calculate several measures, although in the main text we rely on the area under the ROC curve (AUC) (Berge and Jordà (2011)). This curve maps out all possible combinations of type I errors (missed recessions) and type II errors (false alarms). The AUC provides a convenient and easily interpretable summary measure of the indicator's signalling quality.¹⁸ A completely uninformative indicator has an AUC of 0.5 – like tossing a coin; and a perfect one an AUC of 1 – all recessions predicted, no false alarms. The AUC of an informative indicator falls in between and is statistically different from 0.5.

To ensure comparability with the existing literature, we also report other standard measures, such as the mean absolute error (MAE), the root mean square error (RMSE) and the log probability score (LPS) (e.g. Rudebusch and Williams (2009)) and pseudo R-square (e.g. Estrella and Mishkin (1998)). As the main insights do not change, we report these in Annex Table A2.1.

We perform both in-sample and out-of-sample exercises. The in-sample estimation sheds light on the tightness of the link between the variables and recessions with the benefit of hindsight (ex post); the out-of-sample analysis evaluates their performance in real time, i.e. taking into account only the information available up to that point (with the caveat that the latest data vintage instead of the real-time release is used). The latter is a more stringent and useful test, since it replicates the information available to policymakers when they form a judgment on the risks. Because of data limitations, we limit the out-of-sample exercise to advanced economies.

We look at several explanatory variables.

The term spread. We define it as the difference between the 10-year government bond rate and the three-month money market rate. The interest rates are from Bloomberg.

The composite financial cycle indicator. As discussed above, we follow Drehmann et al. (2012).¹⁹ As such, we use bandpass filters with frequencies from eight to 32 years to extract medium-term cyclical fluctuations in real credit, the credit-to-GDP ratio and real property prices, which are then averaged to derive a composite measure of the financial cycle.²⁰ We take the data for total credit, credit-to-GDP ratios and long-run property prices from BIS statistics.

The debt service ratio. This measures interest payment and amortisation of the private non-financial sector relative to income. The BIS publishes data on debt service ratios from 1999 onwards (Drehmann et al. (2015)). We extend them backwards using the same methodology.

As all these variables differ in terms of levels and volatility across countries, we standardise them by their mean and standard

¹⁵ The term spread is constraining the sample for Australia (1986), Finland (1988) and Norway (1988). The DSR is constraining the sample for Ireland (2002).

¹⁶ Brazil, the Czech Republic, Hungary, Korea, Malaysia, Poland, Russia, South Africa and Thailand. For these countries, we always date business cycles using a business cycle-dating algorithm. For an EME to be included, we require that all three variables are available and that we have at least 10 years of spread and DSR data. In addition, we only include an EME if we observe at least one recession in the sample.

¹⁷ The sample is homogeneous for regressions with the DSR and the term spread.

¹⁸ A probability can be transformed into a binary indicator, which is equal to 1 (“on”) if it is above a critical threshold T , and zero (“off”) otherwise. A type I error (missed call) occurs if the binary indicator is “off” but a recession follows; and a type II error (false alarm) if it is “on” but no recession follows. By changing the critical threshold T , the fraction of type I and type II errors changes. Technically, there is no direct mapping between the significance of coefficients in the probit equation and the AUC, especially if the probit includes more than one variable. In that case, the probit regression coefficients may be statistically significant, even if their inclusion does not change the AUC much.

¹⁹ We modify of Drehmann et al. (2012) slightly by applying the filter directly to the log level of the individual series instead of filtering the growth rates.

²⁰ In-sample, it may be a concern that the filter artificially generate forecast performance as the trend in the run-up of recessions is incorporating the lower values during it, thus boosting the gap ahead of it. For us, this is less of an issue for several reasons. First, the DSR is not a filtered series. Second, when we compare short term filtered cycles to simple annual growth rates in Annex 1, the performance of both is equally poor (see Table A1.1). Third, this issue does not arise in the out-of-sample exercise, where the composite financial cycle continues to outperform the term spread (Section 5). This is an important difference to medium term cycles in real GDP. While real-time medium term fluctuations in the real GDP obtained from the same filter are significant predictors in sample, out-of-sample they do not deliver significant AUCs above 0.5.

deviation.²¹

4. In-sample results

As much of the literature focuses on the United States, we present results for this country first and then consider the panel of countries.

It is already evident from the raw data that the term spread provides useful information about the likelihood that the US economy will be in a recession (Annex Graph A2.1, left-hand panel). Unsurprisingly, and as expected from the literature, across all horizons the estimated coefficient is highly statistically significant and negative (Table 1). And the AUCs are very high, up to 0.93 for a one-year horizon (Graph 3 and Table A2.1 upper panel).

That said, the financial cycle indicators are also powerful, in particular the DSR. For a one-year forecast horizon, the DSR actually outperforms the term spread slightly (AUC of 0.96)²² and is insignificantly different from 1 – the perfect indicator (Graph 3 and Table A2.1 upper panel). Again, this is evident from the raw data (Annex Graph A2.1, right-hand panel).

Combining the term spread and financial cycle proxies boosts forecast performance at all horizons. But the gain is very limited in comparison with the spread alone.

The results for the United States should be taken with a large pinch of salt, however. As our sample begins in 1985, we observe only three recessions, with a total of 14 quarters when the economy was in a recession. Hence the need to consider the evidence for a broader set of countries.

The in-sample results for the panel confirm that the financial cycle measures provide valuable information for recessions.

This conclusion is evident for advanced economies, for which a richer sample is available, and regardless of the forecasting horizon. Coefficients for the composite financial cycle measure or the debt service burden are always highly statistically significant (Table 2, upper panel)²³.

In comparison, the term spread seems to be useful only for evaluating recession risk one and two years ahead. The coefficients and AUCs are not statistically significant at the three-year horizon (Table 2, upper panel; Graph 4, left-hand panel; and Table A2.1 middle panel).

Moreover, even at one- and two-year horizons, the composite financial cycle and DSR outperform the term spread. Their AUCs are higher, and the difference is statistically significant (Table A2.1 middle panel).

That said, the financial cycle proxies and the term spread seem to provide complementary information. When they are included jointly in a probit model, they all remain statistically significant up to a two-year horizon. However, the improvement in the AUCs and other evaluation metrics by adding the term spread to financial cycle measures is very limited (Table 2 upper panel; Graph 4 left-hand panel; and Table A2.1 middle panel).

As noted above, while the generally weaker performance of the term spread in the panel may come as a surprise, it reflects, at least in part, the role of the credit risk premium in the yield curve outside the United States. In particular, when we remove the euro area periphery countries (i.e. Ireland, Italy and Spain) from the panel of advanced economies, the AUCs of the term spread increase to 0.71 and 0.65 at the one- and two-year forecasting horizons, respectively. Even then, however, the term spread continues to have less forecasting power than the composite financial cycle indicator or the DSR, whose information content is little affected.

The strong performance of the financial cycle proxies is all the more remarkable given that the methodology stacks the deck against finding significant predictive power. Since the financial cycle, as defined here, tends to be longer than the business cycle, we cannot expect booms to precede all recessions – there will be misses. And because the financial cycle builds up and recedes slowly (e.g. Graph 1), it is likely to sound several “false alarms” before and after the recession has ended. Hence, from the start, the benchmark for the AUC cannot be expected to be 1, i.e. the AUC of the perfect indicator.

The results for EMEs broadly mirror those for advanced economies. Financial cycle proxies are informative, albeit less so than in advanced economies. Coefficients and AUCs are always statistically significant, regardless of forecast horizon. The exception is the DSR, which yields the highest AUC at the one-year horizon but is not significant at the two-year one (Table 2 lower panel; Graph 4 right-hand panel and Table A2.1 lower panel). Despite less developed bond markets, term spreads also seem to provide some valuable information at the one-year horizon (Table 2 lower panel). This is in line with the finding by Mehl (2009).

5. Out-of-sample results

We perform two exercises to assess the indicators’ performance in real time from 1995 onwards: we first examine the effect of (quasi) real-time data, and then the combined effect of (quasi) real-time data and model parameters estimated recursively, i.e. by adding one observation at a time.

To account for the fact that turning points are not known in real time, we assume a fixed lag of four quarters for the turning point

²¹ This has the additional benefit of implicitly controlling for fixed effects in a similar way to Chamberlain’s random effects probit model.

²² The AUCs of the DSR and the term spread are not significantly different statistically.

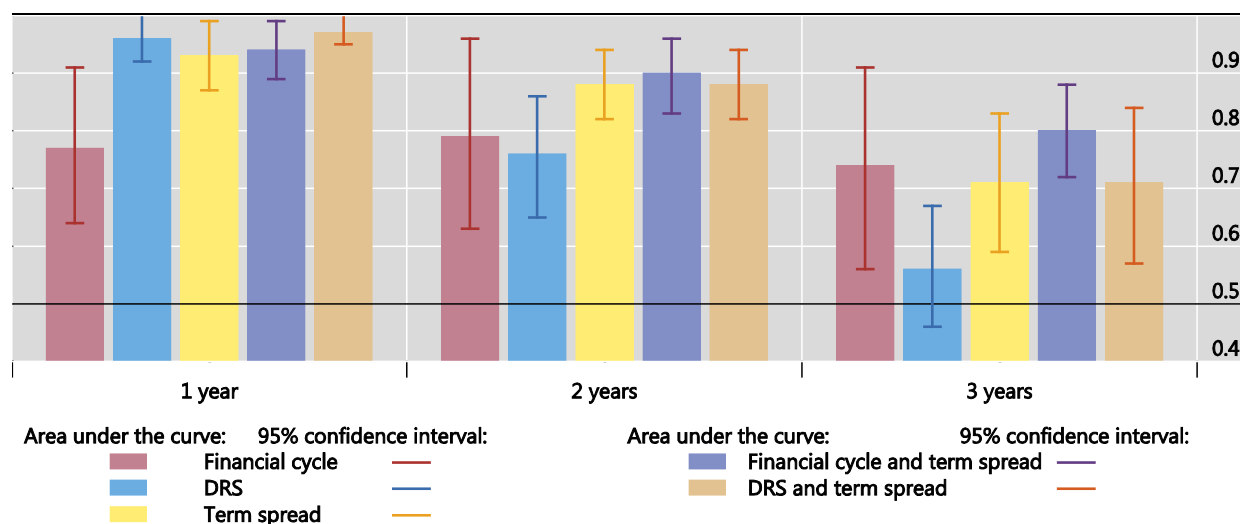
²³ We obtain standard errors by block-bootstrapping in order to account for cross-country correlations.

Table 1Predicting that the economy is in a recession h periods ahead: United States

Regression coefficients from probit models

Horizon		Financial cycle	DSR	Term spread	Financial cycle and term spread	DSR and term spread
1 year	Financial cycle	0.51***			0.29*	
	DSR		2.38***			1.81***
	Spread			-1.51***	-1.41***	-1.14***
2 years	Financial cycle	0.53***			0.33*	
	DSR		0.59***			0.02
	Spread			-1.04***	-0.91***	-1.02***
3 years	Financial cycle	0.43**			0.36**	
	DSR		0.14			-0.14
	Spread			-0.41***	-0.33**	-0.48**

The table reports estimated coefficients from probit regressions (equation (1)) for the United States; ***/**/* indicates significance at the 1/5/10% level.



The horizontal lines at 0.5 indicate the area under the curve of an uninformative, random variable.

Graph 3. AUCs for different forecast horizons: United States

The horizontal lines at 0.5 indicate the area under the curve of an uninformative, random variable.

announcements.²⁴ Ideally, we would use actual lags as in Hwang (2019) who predicts US recessions.²⁵ However, the information is not available for the recession dates from the ECRI. We choose a fixed lag of four quarters as this is close to the average lag for the US (around 14 months since 1985). This assumption also allows for a release lag of GDP series of several months for the countries whose recession dates are determined by the dating algorithm, as the algorithm itself dates turning points with a lag of two quarters (Harding and Pagan (2002)).

We can use only quasi-real-time data, as real-time releases are not available for the panel we look at. But the effects of this may be limited. For one, we already account for a one year lag because of the assumption on the delay in turning point announcements. Moreover, for the spread, real-time data are not an issue, as market data are not revised. In addition, in many countries credit data are also not revised substantially (Drehmann and Tsatsaronis (2014)). Nonetheless, real-time data will affect the DSR and the composite financial cycle indicator through revisions to income, GDP and property prices. As such, our quasi-real-time data only account for the changes in the mean and standard deviation by which variables are normalised. Moreover, we calculate the composite financial cycle indicator recursively, only accounting for information up to that quarter.

To assess the impact of (quasi) real-time data only, we estimate the models up to Q1 1995 and keep the parameters fixed when

²⁴ Specifically, we do not include the last four data points for each regressions.

²⁵ Hwang (2019) not only considers actual lags of turning points announcements but also uses a nowcasting model to predict recession indicators for the periods that the information is unknown at the time of forecasting.

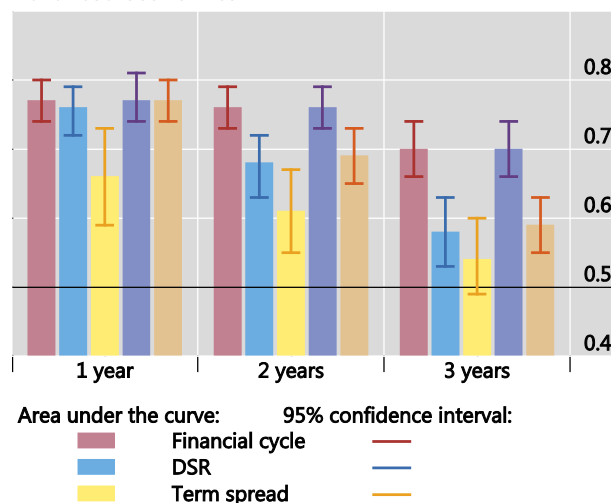
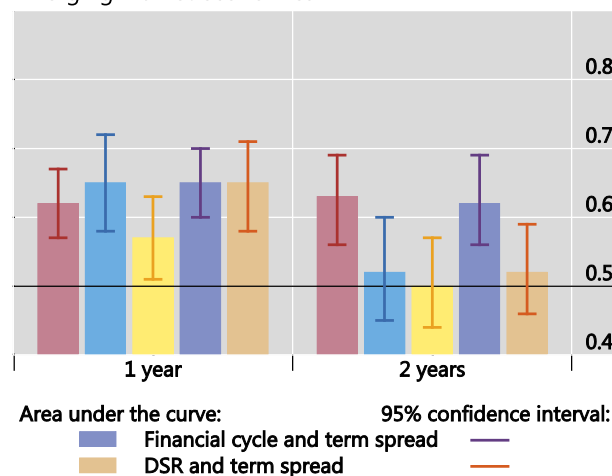
Table 2Predicting that the economy is in a recession h periods ahead: global

Regression coefficients from panel probit models

Horizon		Financial cycle	DSR	Term spread	Financial cycle and term spread	DSR and term spread
Advanced economies						
1 year	Financial cycle	0.69***			0.63***	
	DSR		0.61***			0.57***
	Spread			-0.34***	-0.21***	-0.27***
2 years	Financial cycle	0.63***			0.60***	
	DSR		0.38***			0.35***
	Spread			-0.23***	-0.09	-0.17***
3 years	Financial cycle	0.43***			0.44***	
	DSR		0.16***			0.15***
	Spread			-0.08	0.03	-0.06
Emerging market economies ¹						
1 year	Financial cycle	0.24***			0.25***	
	DSR		0.25***			0.23***
	Spread			-0.12**	-0.17**	-0.03
2 years	Financial cycle	0.25***			0.25***	
	DSR		0.01			0.04
	Spread			0.06	0.03	0.08

The table reports estimated coefficients from probit regressions (equation (1)) for advanced economies and emerging market economies, respectively; ***/**/* indicates significance at the 1/5/10% level.

¹No three-year forecast horizons for EMEs, due to the limited sample.

Advanced economies**Emerging market economies¹**

The horizontal lines at 0.5 indicate the area under the curve of an uninformative, random variable.

¹ No three-year forecast horizons for EMEs, due to the limited sample.

Graph 4. AUCs for different forecast horizons: global

The horizontal lines at 0.5 indicate the area under the curve of an uninformative, random variable.

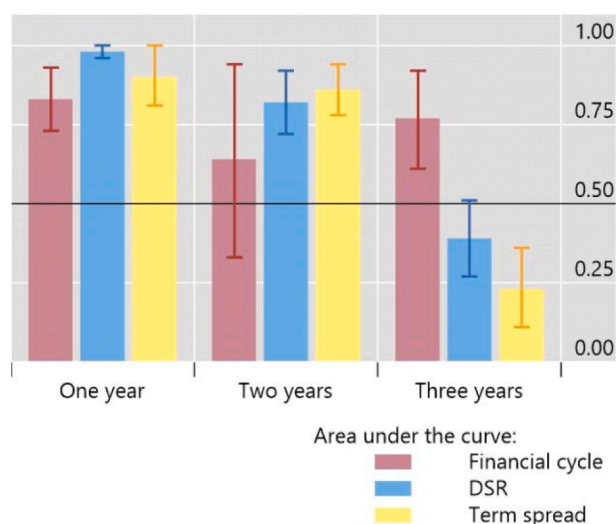
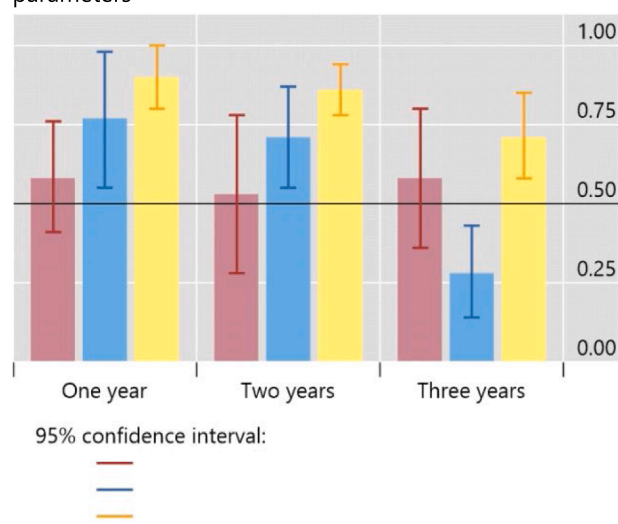
¹No three-year forecast horizons for EMEs, due to the limited sample.

forecasting.

To assess the combined effect of real-time data and time-varying model parameters, we re-estimate the models each period a forecast is made (again starting from Q1 1995) and use the resulting coefficients and real-time data to forecast.

For both cases, thus, the out-of-sample forecast performance is based on the sample from Q1 1995 to Q4 2017.

As above, we first discuss the United States before turning to the panel. We do not consider the specifications with multiple predictors in the case of the United States because we end up with a very limited sample.

Real-time data²Real-time data and recursively estimated model parameters³

The horizontal lines at 0.5 indicate the area under the curve of an uninformative, random variable.

¹ We start the forecasting exercise in Q1 1995. ² Forecasts are calculated using real-time data and fixed parameters estimated with data up to Q1 1995. ³ Forecasts are calculated using real-time data and recursively estimated model parameters, ie by adding one observation at a time.

Graph 5. Out-of-sample AUCs: United States¹

The horizontal lines at 0.5 indicate the area under the curve of an uninformative, random variable.

Even using real-time data, the forecast performance of the financial cycle proxies in the United States is very strong, in particular that of the DSR up to two-year horizon (Graph 5, left-hand panel and Table A2.2). The AUC for a one-year horizon is 0.98 and remains large and significant also at the two-year horizon (0.82).²⁶ The term spread also performs well out of sample up to two-year horizon, as its out-of-sample forecast performance is similar to the full-sample one. The robustness of the US results is remarkable given that the training sample contains only one recession. That said, the sample is even more limited than for the full-sample US results, naturally raising questions about robustness.

The out-of-sample panel results for the financial cycle measures mirror the insights for the United States.²⁷ They continue to have high forecasting power and deliver AUCs statistically significantly higher than 0.5 (Graph 6, left-hand panel and Table A2.3 upper panel). The AUC of the DSR is above 0.6 up to the two-year horizon. While AUCs for the composite financial cycle are lower than in the full-sample estimation, their performance is still statistically significant.

In contrast, the AUCs of the term spread are not statistically significant beyond the one-year horizon. The signal is statistically indistinguishable from that of an uninformative indicator. Even at one-year horizon, AUC of the term spread is significantly lower than that of the DSR. (Graph 6, left-hand panel and Table A2.3 upper panel) The contrast to the results for the United States, confirming that credit risk premia are the key factor behind this result.

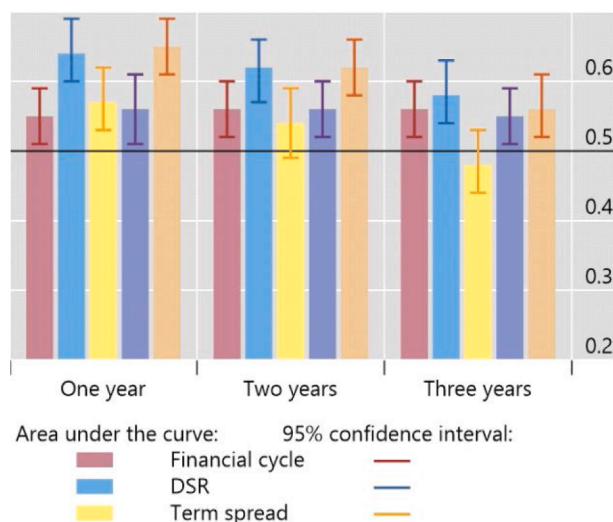
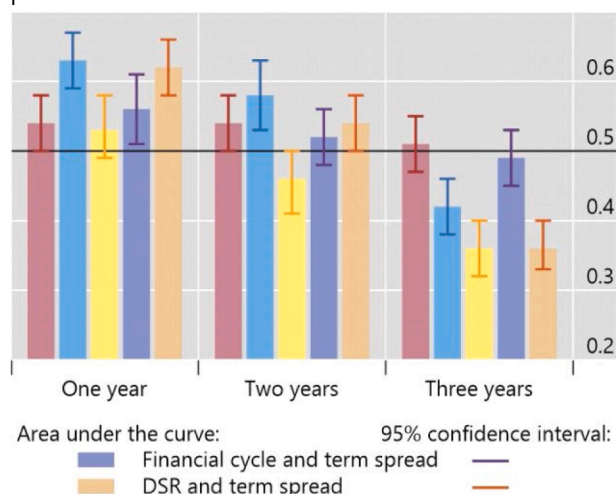
Allowing for changes in model parameters does not alter the broad message (Graph 6, right-hand panel and Table A2.3 lower panel). Both financial cycle proxies deliver AUCs that are statistically higher than 0.5 up to a two-year horizon, although not beyond. Interestingly, allowing parameter estimates to vary as the sample size increases seems to reduce forecast accuracy of the financial cycles to some extent: the AUCs obtained when parameters are fixed are slightly higher.

In comparison, AUCs from the term spread do not differ from that of a random indicator in year one. For year two and three, AUCs are significantly less than 0.5, implying that the probabilities derived from the real time probit model would have been a contrarian signal, i.e. low probabilities would have predicated higher recession risk.²⁸

²⁶ In fact, using the subsample before 1995, the DSR predicts recessions at a one-year horizon perfectly (with infinite β_h in the probit model or equivalently a step function).

²⁷ Dropping individual countries does not significantly change the AUCs for the out-of-sample results. The only exception is that, for a one-year horizon, the AUC of the DSR increases to 0.72 when Japan is dropped.

²⁸ The reason for this is time-varying coefficients, which not only vary in magnitude but also in sign over the sample. Time-varying coefficients are in line with previous findings in the literature for the United States (e.g. Hwang (2019)).

Real-time data²Real-time data and recursively estimated model parameters³

The horizontal lines at 0.5 indicate the area under the curve of an uninformative, random variable.

¹ We start the forecasting exercise in Q1 1995. ² Forecasts are calculated using real-time data and fixed parameters estimated with data up to Q1 1995. ³ Forecasts are calculated using real-time data and recursively estimated model parameters, i.e. by adding one observation at a time.

Graph 6. Out-of-sample AUCs: advanced economies¹

The horizontal lines at 0.5 indicate the area under the curve of an uninformative, random variable.

¹We start the forecasting exercise in Q1 1995. ²Forecasts are calculated using real-time data and fixed parameters estimated with data up to Q1 1995. ³Forecasts are calculated using real-time data and recursively estimated model parameters, i.e. by adding one observation at a time.

6. Predicting turning points

For robustness, we look at the forecasting power of the financial cycle and the term spread for assessing the likelihood that the cycle will turn from boom to bust (i.e. a recession starts) within the next one, two or three years. In comparison with our analysis in the main text, this forecasting exercise differs along two key dimensions.

First, it focuses on pre-recession periods only. In the previous exercise, when we estimate the likelihood that the economy will be in a recession, the sample contains many observations for which the machine is “predicting” a recession *when the economy is already in one*. For instance, for a one-year prediction horizon, in our sample 50% of the observations followed by a recession in four quarters are times when the economy is already in a recession. While this is standard in the literature, it is of less value for policymakers and practitioners.

Second, this definition does not impose a strict lead-lag relationship on the data. Instead, the interest is on whether a recession will begin within a given interval of time.

To operationalise this recession indicator, one question is how to treat recession periods. Following the early warning indicator literature for financial crises, we drop them. The reason is that setting the indicator to 0 during this interval would bias the results (Borio and Lowe (2002), Bussière and Fratzscher (2006)).

As the three-year forecasting window is quite long, we are also interested to see to which year the forecasting power applies.²⁹ Hence, we consider one, two and three years ahead separately. This in turn raises the question of how to treat observations within the three-year forecast horizon of interest but outside the exact estimation window. In other words, how should one treat economically correct signals (e.g. a negative spread) that are issued within the three-year horizon (e.g. two years ahead) but outside the window of interest (e.g. whether the recession will start the following year)? In our earlier analysis, these signals would be considered “wrong” given the strict lead-lag relationship, despite being correct economically. To overcome this problem, we follow Drehmann and Juselius (2014) and ignore the informational content of the variables within the three-year horizon but not within the one-, two- or three-year horizon considered.

Accordingly, we define an indicator variable $I_{i, h, t} = 1$ one, two or three years before the beginning of a recession. We do not consider information during recessions and in the two years before recessions that are outside the estimation horizon of interest: in all other

²⁹ We also ran exercises with the whole three-year forecasting horizon. The results again confirm the message from the main analysis. The results are available upon request.

periods, $I_{i,h,t} = 1$, $t=0$.

To clarify the differences between the two different recession risk measures, Annex Table A2.2 provides an example of the two approaches for different horizons around the recession in the United States starting in 2007.

Similarly to the main analysis, we use a panel probit model to predict the probability of a recession starting within the next h periods. Specifically, we have:

$$\text{Prob}(I_{i,h,t} = 1 | X_{i,t}) = \Phi(\alpha_h + \beta_h' X_{i,t}) \quad (2)$$

Both in-sample and out-of-sample results confirm the message of the main analysis. Financial cycle proxies contain valuable information in evaluating recession risks; and they tend to outperform the term spread (Annex Table A2.5-7 and Graph A2.2).

Interestingly, across the variables and the different forecast horizons, there are no major differences between the AUCs for predicting whether the recession will start any time within a given horizon or whether the economy will be in recession in a specific quarter ahead. To be sure, point estimates differ. For instance, the full-sample AUC of the financial cycle is higher and that of the DSR lower when predicting the turning point. Yet confidence bands generally overlap, so that differences are not statistically significant. This is reassuring, as most practitioners and commentators use the results of the literature mainly when the economy is still in boom and not yet in a recession.

7. Conclusion

Business cycles may not die of old age (Rudebusch (2016)), but if financial booms develop, they become more fragile. This is the case in both advanced economies and emerging market economies. Moreover, given that financial cycles build up slowly, the corresponding proxies provide information about recession risk even at a three-year horizon. And when we run a horse race against the term spread – the indicator most widely used to assess recession risk – we find that they outperform the term spread in both in-sample and out-of-sample exercises. The debt service ratio is particularly effective in this respect. These results suggest that financial cycle proxies may be another indicator that could be useful to policymakers, professional forecasters and market participants more generally.

Table A1.1

CRedit authorship contribution statement

Claudio Borio: Conceptualization, Writing - original draft. **Mathias Drehmann:** Conceptualization, Methodology, Writing - original draft. **Fan Dora Xia:** Methodology, Formal analysis, Writing - original draft.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jmacro.2020.103258](https://doi.org/10.1016/j.jmacro.2020.103258).

Annex 1: Alternative financial cycle measures

In this Annex we examine the forecasting performance of credit and property price developments separately. And we consider their short and medium-term cycles as well as growth rates.

The medium-term cycles for real credit, real property prices and credit to GDP are the individual input series into the composite financial cycle. As discussed in the main text, medium-term cycles are derived by a bandpass filter with frequencies from eight to 30 years.³⁰ The short term cycles for the three series are also derived with a band-pass filter. But we extract standard business cycle frequencies between 6 and 32 quarters. As a filter free benchmark, we also look at the 1 year growth rates in real credit, real property prices and the credit-to-GDP ratio.

Several results stand out.

First, medium-cycles in credit and residential property prices outperform short term cycles (i.e. cycles at business cycle frequency) in predicting recessions. For advanced economies, the AUC of medium-term cycle is on average 16 percentage points higher than the AUC of the short term cycle. For EMEs, the difference is less stark but remains 5 percentage points on average. The difference is largest for the medium versus short term credit cycle over a one year horizon in advanced economies. Whereas the medium-term credit cycle has a highly significant AUC of 0.77, the AUC of the short term credit cycle is 0.53 and as such has no different forecasting power than a coin toss. Compared to the term spread, medium-term cycles always have higher AUCs. In most cases, this difference is significant at 5% confidence level. In contrast, short term cycles perform significantly worse than the term spread in several cases.

Second, the composite financial cycle captures both the superior performance of the medium-term credit cycle for short (i.e. 1 year) and that of the medium-term property prices cycle for longer forecast horizons (i.e. 3 years). The composite financial cycle AUC is

³⁰ In the working paper version of this paper, we also look at alternative measures of medium credit and property prices, namely the deviations of credit-to-GDP \ property prices from a one-sided Hodrick-Prescott (HP) filtered trend with a smoothing parameter, λ , of 400,000 as well as the three year growth rate in the credit-to-GDP ratio as a popular measure in literature (e.g. Mian et al. (2017) or Jordà et al. (2013)). Results are in line with medium-term, bandpass filtered series.

Table A1.1Predicting that the economy is in a recession in h periods: Alternative financial cycle measures

		term spread	Composite financial cycle	credit annual growth	credit/GDP Short-term cycle	property Medium-term cycle	credit	credit/GDP	property	credit	credit/GDP	property
Advanced economies												
1 year	AUC	0.66	0.77	0.54	0.56	0.62	0.53	0.60	0.62	0.77	0.71	0.74
	Low	0.62	0.74	0.51	0.53	0.58	0.50	0.57	0.58	0.74	0.68	0.71
	Up	0.70	0.80	0.58	0.59	0.65	0.56	0.63	0.65	0.79	0.74	0.77
	Sig-spread	.	0.00	0.00	0.00	0.11	0.00	0.02	0.11	0.00	0.03	0.00
2 years	AUC	0.61	0.76	0.60	0.61	0.57	0.53	0.53	0.64	0.75	0.65	0.76
	Low	0.58	0.73	0.57	0.58	0.54	0.49	0.50	0.60	0.72	0.62	0.73
	Up	0.65	0.79	0.64	0.64	0.60	0.56	0.56	0.67	0.77	0.68	0.79
	Sig-spread	.	0.00	0.90	0.87	0.03	0.00	0.00	0.25	0.00	0.11	0.00
3 years	AUC	0.54	0.70	0.65	0.66	0.63	0.54	0.54	0.51	0.67	0.57	0.73
	Low	0.51	0.67	0.62	0.63	0.60	0.51	0.51	0.47	0.64	0.54	0.70
	Up	0.58	0.73	0.68	0.69	0.66	0.58	0.58	0.55	0.70	0.60	0.76
	Sig-spread	.	0.00	0.00	0.00	0.00	0.94	0.98	0.19	0.00	0.24	0.00
Emerging market economies												
1 year	AUC	0.58	0.62	0.65	0.65	0.58	0.59	0.55	0.72	0.68	0.69	0.59
	Low	0.52	0.56	0.60	0.60	0.52	0.53	0.49	0.67	0.62	0.63	0.53
	Up	0.63	0.68	0.71	0.70	0.63	0.65	0.61	0.78	0.74	0.74	0.65
	Sig-spread	.		0.02	0.04	0.76	0.66	0.61	0.00	0.02	0.01	0.96
2 years	AUC	0.50	0.62	0.66	0.63	0.55	0.56	0.63	0.52	0.64	0.63	0.65
	Low	0.44	0.55	0.60	0.58	0.49	0.51	0.58	0.46	0.58	0.58	0.59
	Up	0.56	0.70	0.72	0.69	0.61	0.61	0.68	0.59	0.70	0.69	0.71
	Sig-spread		0.00	0.00	0.00	0.19	0.10	0.00	0.52	0.00	0.00	0.

In-sample results; Low\Up: lower\upper value of the 95% confidence bands; Sig-spread: value for significance test of whether AUC is different to the term spread AUC.

never higher, but generally as high as the highest AUC of an individual medium-term cycle for a specific horizon. Given the different performance of credit and property medium-term cycles for short and longer horizon this highlights that the composite financial cycle provides a useful tool across all horizons that can be easily interpreted by policy makers and practitioners.³¹

Last, there is no major difference in forecast performance of growth rates and short term cycles. Growth rates tend to perform better but on average the difference between AUCs is just 3.5 percentage points. This shows that it is indeed the medium cyclical frequencies rather than the statistical filter that driver forecast performance.

References

- Adrian, T, Grinberg, F, Liang, N, Malik, S, 2018. The term structure of growth-at-risk. In: *IMF Working Papers*, no WP/18/180, August.
- Aikman, D, Haldane, A, Nelson, B, 2015. Curbing the credit cycle. *The Economic Journal* 125 (585), 1072–1109.
- Bank for International Settlements (2018): *Annual Economic Report 2018*, Chapter I, June.
- Baro, C., 2016. Monetary and prudential policies at a crossroads? New challenges in the new century. In: *BIS Working Papers*, no 216, September.
- Berge, TJ, 2015. Predicting recessions with leading indicators: Model averaging and selection over the business cycle. *Journal of Forecasting* 34 (6), 455–471.
- Berge, TJ, Jordà, O, 2011. Evaluating the classification of economic activity into recessions and expansions. *American Economic Journal: Macroeconomics* 3 (2), 246–277.
- Bernanke, B, Gertler, M, Gilchrist, S, 1999. The financial accelerator in a quantitative business cycle framework. *Handbook of macroeconomics* 1, 1341–1393.
- Borio, C, 2014. The financial cycle and macroeconomics: what have we learnt? *Journal of Banking & Finance* 45, 182–198. August. Also available as *BIS Working Papers*, no 395, December 2012.
- Borio, C, Drehmann, M, 2009. Assessing the risk of banking crises – revisited. *BIS Quarterly Review* 29–46. March.
- Borio, C, Kharroubi, E, Upper, C, Zampolli, F, 2016. Labour reallocation and productivity dynamics: financial causes, real consequences. In: *BIS Working Papers*, no 534, January.
- Borio, C, Lowe, P, 2002. Asset prices, financial and monetary stability: exploring the nexus. In: *BIS Working Papers*, no 114, July.
- Brunnermeier, MK, Sannikov, Y, 2014. A macroeconomic model with a financial sector. *American Economic Review* 104 (2), 379–421.
- Burns, AF, Mitchell, WC, 1946. *Measuring Business Cycles*. National Bureau of Economic Research, New York.
- Bussière, M, Fratzscher, M, 2006. Towards a new early warning system of financial crises. *Journal of International Money and Finance* 25 (6), 953–973.
- Chauvet, M, Potter, S, 2002. Predicting a recession: Evidence from the yield curve in the presence of structural breaks. *Economics Letters* 77 (2), 245–253.
- Claessens, S, Kose, M, 2018. Frontiers of macrofinancial linkages. In: *BIS Papers*, no 95, January.
- Claessens, S, Kose, M, Terrones, M, 2012. How do business and financial cycles interact? *Journal of International Economics* 87 (1), 178–190.
- Coroneo, L and S Pastorello (2017): “European spreads at the interest rate lower bound”, unpublished manuscript.
- de Winter, J, Koopman, S J, Hindrayanto, I, Chouhan, A, 2017. Modeling the business and financial cycle in a multivariate structural time series model. In: *DNB Working Papers* no 537.
- Detken, C, Weeken, O, Alessi, L, Bonfim, D, Boucinha, M, Castro, C, Frontczak, S, Giordana, G, Giese, J, Jahn, N, Kakes, J, Klaus, B, Lang, J-H, Puzanova, N, Welz, P, 2014. Operationalising the countercyclical capital buffer: indicator selection, threshold identification and calibration options. In: *Occasional Papers*, no 5. European Systemic Risk Board.
- Drehmann, M, Borio, C, Tsatsaronis, K, 2012. Characterising the financial cycle: don't lose sight of the medium term!. In: *BIS Working Papers*, no 380, June.
- Drehmann, M, Ilnes, A, Juselius, M, Santos, M, 2015. How much income is used for debt payments? A new database for debt service ratios. *BIS Quarterly Review* 89–103. September.
- Drehmann, M, Juselius, M, 2014. Evaluating early warning indicators of banking crises: satisfying policy requirements. *International Journal of Forecasting* 30 (3), 759–780.
- Drehmann, M, Juselius, M, Korinek, A, 2018. Going with the flows: new borrowing, debt service and the transmission of credit booms. In: *NBER Working Papers*, no 24549.
- Drehmann, M, Tsatsaronis, K, 2014. The credit-to-GDP gap and countercyclical capital buffers: questions and answers. *BIS Quarterly Review* 55–73. March.
- Eichengreen, B, Mitchener, K, 2003. The Great Depression as a credit boom gone wrong. In: Wolcott, S, Hanes, C (Eds.), *Research in Economic History*, 22. Emerald Group Publishing Limited, pp. 183–237.
- Estrella, A, Mishkin, F, 1998. Predicting US recessions: financial variables as leading indicators. *Review of Economics and Statistics* 80 (1), 45–61.
- Fendel, R, Mai, N, Mohr, O, 2018. Recession probabilities for the Eurozone at the zero lower bound. In: *WHU Working Paper Series in Economics*, no 18/04.
- Fisher, I, 1911. The equation of exchange, 1896–1910. *The American Economic Review* 1 (2), 296–305.
- Galati, G, Hindrayanto, I, Koopman, S J, Vlekke, M, 2016. Measuring financial cycles in a model-based analysis: Empirical evidence for the United States and the euro area. *Economics Letters* 145, 83–87.
- Geanakoplos, J, 2010. The leverage cycle. In: Acemoglu, D, Rogoff, K, Woodford, M (Eds.), *NBER Macroeconomics Annual 2009*. University of Chicago Press, Chicago and London.
- Grabowski, S, 2009. The financial indicators leading real economic activity – the case of Poland. *Central European Journal of Economic Modelling and Econometrics* 4 (1), 311–332.
- Guender, A, 2018. Credit prices vs credit quantities as predictors of economic activity in Europe: which tell a better story? *Journal of Macroeconomics* (57), 380–399.
- Harding, D, Pagan, A, 2002. Dissecting the cycle: a methodological investigation. *Journal of Monetary Economics* 49 (2), 365–381.
- Hwang, Y, 2019. Forecasting recessions with time-varying models. *Journal of Macroeconomics* 62, 103–153.
- Huffman, W, Lothian, J, 1984. The gold standard and the transmission of business cycles, 1833–1932. In: Bordo, D, Schwartz, A (Eds.), *A retrospective on the classical gold standard*. NBER, pp. 1821–1931.
- Jordà, O, Schularick, M, Taylor, A, 2013. When credit bites back. *Journal of Money, Credit and Banking* 45 (s2), 3–28.
- Juselius, M, Drehmann, M, 2020. Leverage dynamics and the burden of debt. *Oxford Bulletin of Economics and Statistics* 82 (2), 347–364.
- Kindleberger, C., 2000. *Maniacs, panics and crashes*. Cambridge University Press, Cambridge.
- Kiyotaki, N., Moore, J, 1997. Credit cycles. *Journal of political economy* 105 (2), 211–248.
- Liu, W, Mönch, E, 2016. What predicts US recessions? *International Journal of Forecasting* 32 (4), 1138–1150.
- Mehl, A, 2009. The yield curve as a predictor and emerging economies. *Open Economies Review* 20 (5), 683–716.
- Mian, A, Sufi, A, Verner, E, 2017. Household debt and business cycles worldwide. *The Quarterly Journal of Economics* 132 (4), 1755–1817.
- Mian, A R, Straub, L, Sufi, A, 2020. Indebted demand. In: *NBER Working Papers*, no 26940.

³¹ Estimating the probit regression with all three medium-term cycles separately increases the forecast performance. It generates somewhat higher AUCs but risks overfitting. In addition, practitioners typically value the interpretability of forecasts more than their accuracy (e.g. Önkai et al. (2002)), underscoring the benefit of a single, composite financial cycle indicator.

- Minsky, HP, 1982. Can it happen again? Essays on instability and finance. M E Sharpe, Amonk.
- Ng, E C Y, 2012. Forecasting us recessions with various risk factors and dynamic probit models. *Journal of Macroeconomics* 34 (1), 112–125.
- Ng, S, Wright, J H, 2013. Facts and challenges from the great recession for forecasting and macroeconomic modeling. *Journal of Economic Literature* 51 (4), 1120–1154.
- Oman, W, 2019. The synchronization of business cycles and financial cycles in the euro area. *International Journal of Central Banking* 15 (1), 327–362.
- Önköl, D, Thomson, ME, Pollock, A A C, 2002. Judgmental forecasting. In: Clements, M P, Hendry, D F (Eds.), *A companion to economic forecasting*. Blackwell Publishers, Malden and Oxford.
- Öztürk, H, Pereira, L, 2013. Yield curve as a predictor of recessions: evidence from panel data. *Emerging Markets Finance and Trade* 49, 194–212.
- Paweenawat, A, 2017. The information content of the term structure of interest rates in emerging economies: the case of Thailand. *Journal of Emerging Market Finance* 16 (2), 136–150.
- Ponka, H, 2017. The role of credit in predicting US recessions. *Journal of Forecasting* 36 (5), 469–482.
- Rudebusch, G, 2016. Will the economic recovery die of old age? *Economic Letters*. Federal Reserve Bank of San Francisco, February.
- Rudebusch, G, Williams, J, 2009. Forecasting recessions: the puzzle of the enduring power of the yield curve. *Journal of Business and Economic Statistics* 27 (4), 492–503.
- Runstler, G, Vlekke, M, 2018. Business, housing, and credit cycles. *Journal of Applied Econometrics* 33 (2), 212–226.
- Schuler, Y S, Hiebert, P P, Peltonen, T A, 2020. Financial cycles: Characterisation and real-time measurement. *Journal of International Money and Finance* 100.
- Schularick, M, Taylor, A, 2012. Credit booms gone bust: monetary policy, leverage cycles, and financial crises, 1870–2008. *American Economic Review* 102 (2), 1029–1061.
- Strohsal, T, Proano, C R, Wolters, J, 2019. Characterizing the financial cycle: Evidence from a frequency domain analysis. *Journal of Banking and Finance* 106, 568–591.
- Verona, F, 2016. Time–frequency characterization of the U.S. financial cycle. *Economics Letters* 144, 75–79.
- Zarnowitz, V, 1999. Theory and history behind business cycles: are the 1990s the onset of a golden age? *Journal of Economic Perspectives* 13 (2), 69–90.
- Zulkhibri, M, Rani, M, 2016. Term spread, inflation and economic growth in emerging markets: evidence from Malaysia. *Review of Accounting and Finance* 15 (3), 372–392.